

Lecture 7

Posterior Predictive Distribution

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(Edited from the lecture notes by Brian J. Reich, NC State)



Bayesian prediction

- ▶ Often the objective is to predict a future event.
- ▶ Example: Last spring we planted $n = 10$ seedlings and $Y = 2$ survived the winter. If we plant $n = 10$ plants again this year, what is the probability at least one will survive the winter?
- ▶ Let Y^* be the predicted value and θ be the true survival probability.
- ▶ If the parameters were known, then we would predict

$$Y^*|\theta \sim \text{Binomial}(10, \theta),$$

and thus, $\text{Prob}(Y > 0) = 1 - (1 - \theta)^{10}$.

- ▶ Of course, if we knew the parameters, we would be doing probability and not statistics.

Two types of uncertainty

- ▶ Sampling: Even if we knew θ , we could not predict Y exactly because there is inherent randomness in which plants survive. This is quantified using the likelihood distribution, $f(Y|\theta)$.
- ▶ Parametric: We can never know θ exactly. This uncertainty is quantified by its prior and posterior distributions.
- ▶ How to resolve each type of uncertainty?

Plug-in approach

- ▶ One approach to prediction is the “plug-in” approach.
- ▶ That is, if $\hat{\theta}$ is an estimate, then use prediction distribution

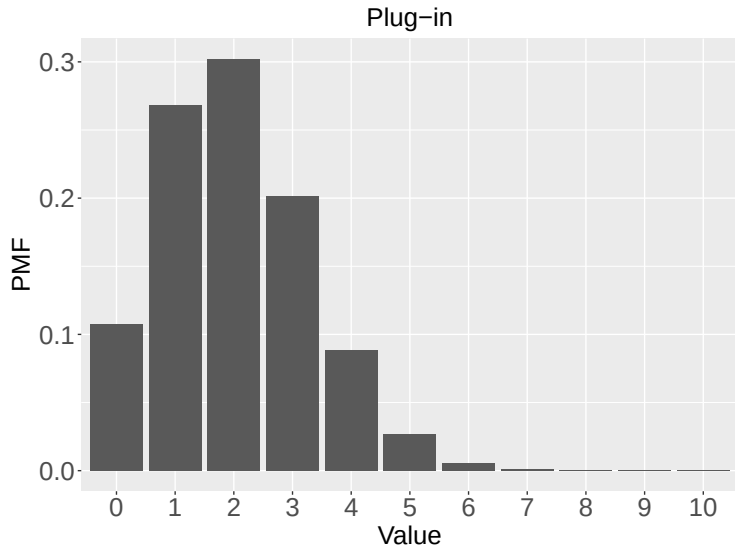
$$Y^* \sim f(Y|\hat{\theta}).$$

- ▶ Example: Here $\hat{\theta} = 2/10$ and then we calculate

$$\text{Prob}(Y > 0) = 1 - (1 - 0.2)^{10}.$$

- ▶ If $\hat{\theta}$ has small uncertainty, then this is fine.
- ▶ Otherwise, this underestimates uncertainty in Y^* .

Plug-in approach



The PPD

- ▶ For the sake of prediction, the parameters are not of interest.
- ▶ They are vehicles by which the data inform about the predictive model.
- ▶ The **Posterior Predictive Distribution** (PPD) averages over their posterior uncertainty

$$f(Y^*|Y) = \int f(Y^*|\theta)p(\theta|Y)d\theta.$$

- ▶ This properly accounts for parametric uncertainty.
- ▶ The input is data, and the output is a prediction distribution.

An example of PPD

- Suppose, our model is

Likelihood: $Y|\theta \sim \text{Binomial}(N, \theta)$,

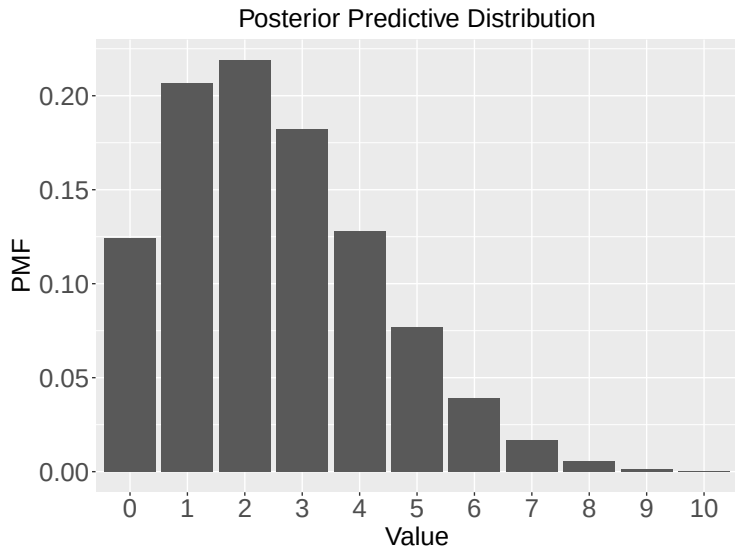
Prior: $\theta \sim \text{Beta}(a, b)$.

- We saw that the posterior is then

$$\theta|Y \sim \text{Beta}(Y + a, N - Y + b).$$

- The PPD is

$$\begin{aligned} f(Y^*|Y) &= \int f(Y^*|\theta)p(\theta|Y)d\theta \\ &= \int \binom{N}{Y^*} \theta^{Y^*} (1 - \theta)^{N-Y^*} \frac{\Gamma(N + a + b)}{\Gamma(Y + a)\Gamma(N - Y + b)} \theta^{Y+a-1} (1 - \theta)^{N-Y+b-1} d\theta \\ &= \binom{N}{Y^*} \frac{\Gamma(N + a + b)}{\Gamma(Y + a)\Gamma(N - Y + b)} \frac{\Gamma(Y + Y^* + a)\Gamma(2N - Y - Y^* + b)}{\Gamma(2N + a + b)}. \end{aligned}$$



Computing the PPD

- ▶ Monte Carlo sampling approximates the PPD.
- ▶ Say $\theta^{(1)}, \dots, \theta^{(S)}$ are samples from the posterior.
- ▶ If we make a sample for Y^* for each $\theta^{(s)}$,

$$Y^{*(s)} \sim f(Y|\theta^{(s)}),$$

then the $Y^{*(s)}$ are samples from the PPD.

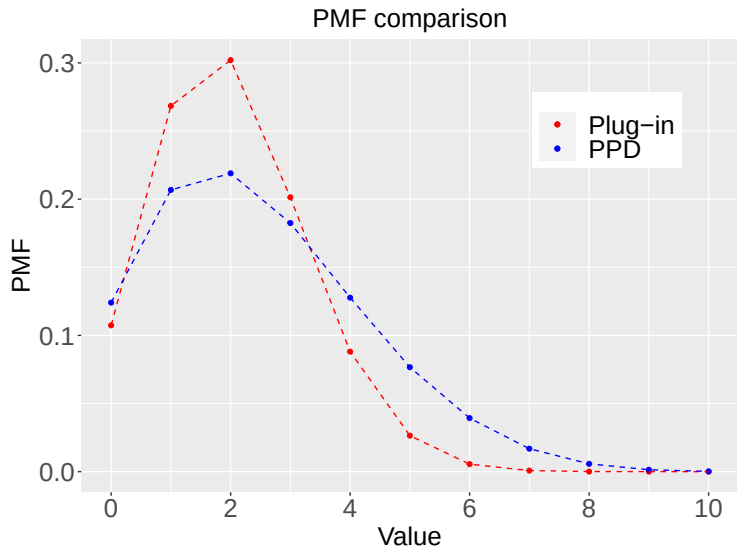
- ▶ The posterior predictive mean is approximated by the sample mean of the $Y^{*(s)}$.
- ▶ The probability that $Y^* > 0$ is approximated by the sample proportion of the $Y^{*(s)}$ that are non-zero.

Bayesian prediction

```
> # Data
> Y <- 2; n <- 10

> # The posterior is  $\theta|Y \sim \text{Beta}(A, B)$ 
> A <- Y+1; B <- n-Y+1
>
> # Plug in estimate of  $P(Y_{\text{star}} > 0)$ 
> 1-dbinom(0,10,.2)
[1] 0.8926258
>
> # Approximate the PPD using MC sampling
> theta <- rbeta(100000,A,B)
> Ystar <- rbinom(100000,10,theta)
> mean(Ystar>0)
[1] 0.87454
```

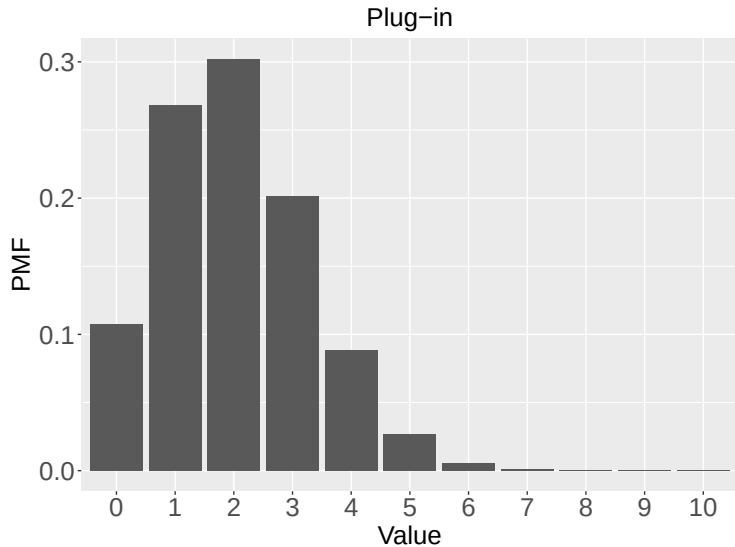
Plug-in versus PPD

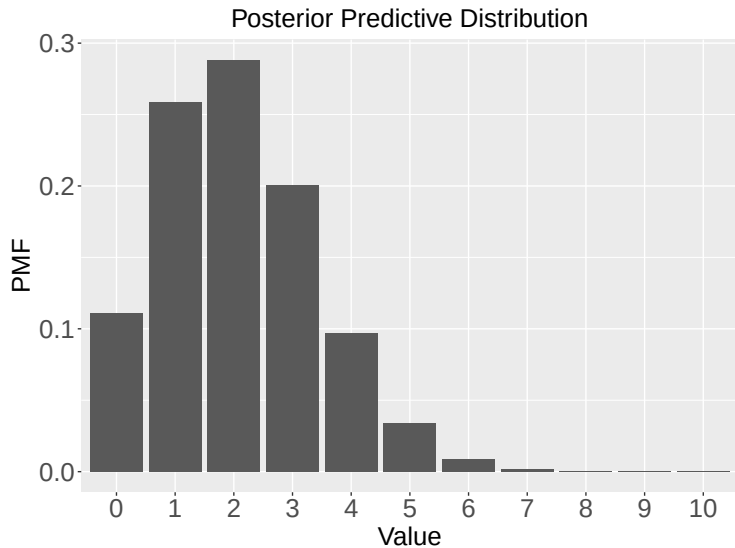


Another example

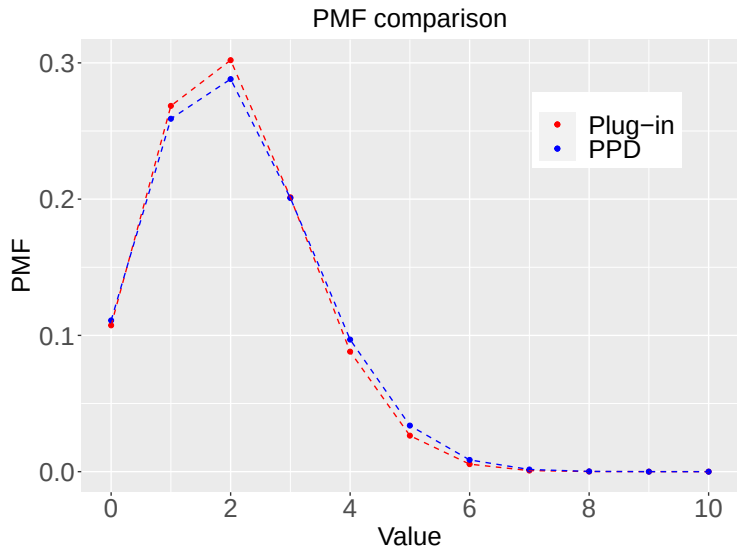
- ▶ Example: Last spring we planted $n = 100$ seedlings and $Y = 20$ survived the winter. If we plant $n = 10$ plants again this year, what is the probability at least one will survive the winter?
- ▶ We again compare the plug-in and the PPD approaches.

Plug-in approach





Plug-in versus PPD



Thank you!